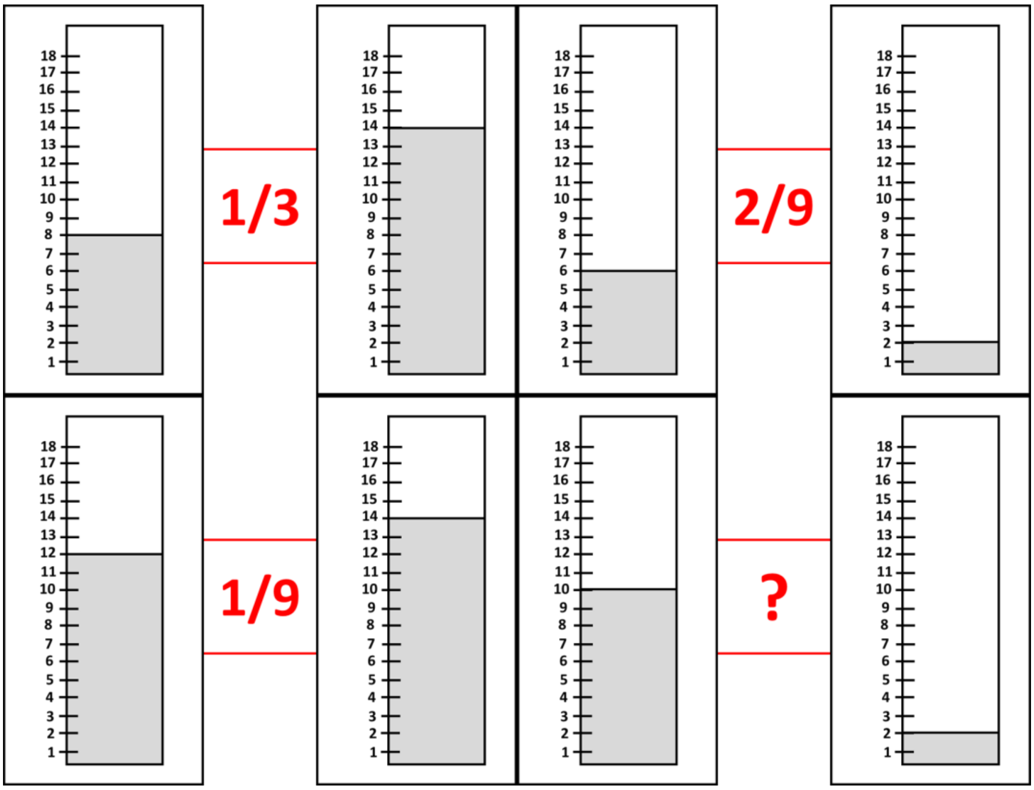
Chapter 8: Working with Fractions

1. **Samuel received chocolates for his birthday. He decided to share it with his family.**
   - He keeps 1/3rd of the chocolates for himself
   - He gives 1/4th of what is left to his sister Sarah
   - He gives the remaining 12 chocolates (except the ones which he kept for himself) to his brother
   How many chocolates had Samuel received in total before he shared it?

   a) 20 b) 22 c) 24 d) None of these

---

2. **If each of the given terms follows the same theme, what will come in place of "?"**



   a) 2/9 b) 2/3 c) 4/9 d) 1/9

---

3. **A bus started its journey from Mumbai.**
   - 1/4th of the passengers got off the bus in Pune and 35 new passengers got on the bus
   - Half of the passengers got off the bus in Goa with no new passengers getting on the bus
   When the bus left from Goa, the total number of passengers was 85. How many passengers boarded the bus from Mumbai?

   a) 135 b) 170
   c) 180 d) Cannot be Determined

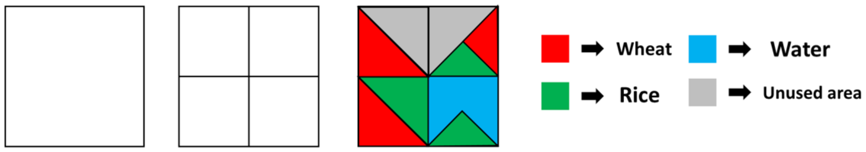
4. **Which of the following statements is/are sufficient to answer the given question?**
Question: What fraction of the initial liquid remains in the vessel after Day 2?
Statement 1: The vessel originally contained 100 litres of liquid
Statement 2: On Day 1, the amount of liquid that evaporates is equal to half of the quantity that would remain if one-third of the initial liquid were removed
Statement 3: On Day 2, the liquid left in the vessel becomes one-fourth of the quantity that was present at the beginning of Day 2

a) Statement 3 alone is sufficient
b) Both Statement 1 and Statement 2 are necessarily required
c) Both Statement 2 and Statement 3 are necessarily required
d) Question cannot be answered even if all the Statements are used

---

5. **A farmer has a square farm as shown in Image 1. He divides it into 4 equal parts as shown in Image 2. Now, he allotted certain portions of land for specific purposes (Rice crop, Wheat crop, Water) and some part of the farm is unused, as shown in Image 3. What fraction of the new farm is occupied by wheat?**

Image 1 Image 2 Image 3



a) 3/16 b) 1/4 c) 5/16 d) 6/16

---

6. **A wire is bent to form a rectangle with length L and breadth B. After some time, the length is increased by one-fifth of its original value, and the breadth is decreased by one-fifth of its original value. After this change:**
   - **The perimeter of the rectangle remains the same**
   - **The area of the rectangle decreases by 100 sq. cm**

What was the original length of the rectangle?

a) 25 cm b) 40 cm c) 45 cm d) 50 cm

---

7. **A number sequence is mentioned below:**
19/2,38/6,76/18,152/54,…
Which will be the first term of the series that will be less than 1?

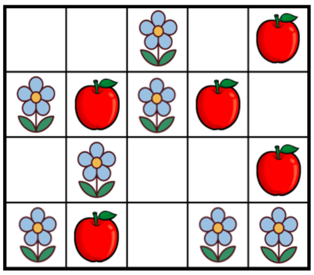
a) 7th term of the series
b) 8th term of the series
c) 10th term of the series
d) None of these

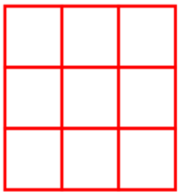
---

8. **A box has 98 chalks. Each day, a teacher picks up a chalk to write.**
When a chalk becomes 1/7 of its original size, she sets it aside.
Whenever she collects enough such small pieces to make one full chalk, she joins them and uses it. For how many days will the 98 chalks last?

a) 84 b) 98 c) 112 d) 114

9. **A grid contains apples and flowers in some of its cells. A 3 × 3 square frame can be placed on the grid without rotating and without extending outside the grid, covering exactly 9 cells. In how many different positions can the frame be placed so that at least one-third of the covered cells contain flowers?**





3 x 3 GRID

a) 3 b) 5 c) 4 d) 6

---

10. **Each circle is divided into equal parts, with some parts shaded. A fraction is written to the right of each circle. How many of these fractions correctly represent the shaded part of their corresponding circles?**

a) 4 b) 3 c) 2 d) 5

The Thinking Spot

You have a collection of 10 different squares arranged as shown below. What is the minimum number of pairs of squares you need to interchange to ensure that white and black squares are arranged alternately?

Note: *If you swap 1 black square with 1 white square, it is counted as 1 PAIR OF SQUARES being swapped*

(a) 1 (b) 2 (c) 3 (d) 5